

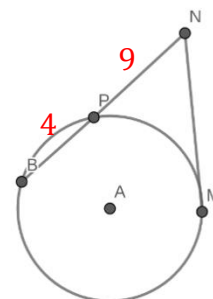
Geometry
Unit 12
Lesson 3 Homework

Name _____
Date _____
Period _____

1. Circle A is shown with points M , B and P on the circle. Secant $\overline{BN}$ and tangent $\overline{MN}$ intersect at point N outside the circle. $PB = 4$ and $PN = 9$. What is the length of $\overline{MN}$? Round to the nearest thousandth.

$$9(13) = (MN)^2$$

$$MN \approx 10.817$$



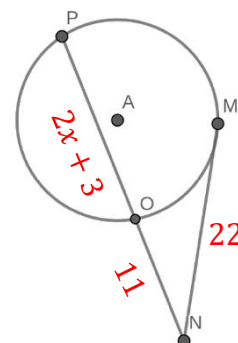
2. Circle A is shown with points M , O and P on the circle. Secant $\overline{PN}$ and tangent $\overline{MN}$ intersect at point N outside the circle. $PO = (2x + 3)$, $ON = 11$ and $MN = 22$. Find the value of x .

$$11(2x + 3 + 11) = 22^2$$

$$22x + 154 = 484$$

$$22x = 330$$

$$x = 15$$



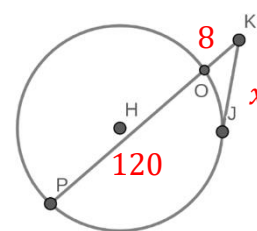
3. What is the length of $\overline{PN}$?

$$2(15) + 3 + 11 = PN = 44$$

4. Circle H is shown with points J , O and P on the circle. Secant $\overline{KP}$ and tangent $\overline{KJ}$ intersect at point K outside the circle. $PO = 120$, $OK = 8$ and $KJ = x$. Find the value of x .

$$8(128) = x^2$$

$$x = 32$$



5. Using the same image as question 4, circle H is shown with points J , O , and P on the circle. Secant $\overline{KP}$ and tangent $\overline{KJ}$ intersect at point K outside the circle. $PO = 12 + x$, $OK = 10$ and $KJ = 18$. Find the length of $\overline{KP}$.

$$10((12 + x) + 10) = 18^2$$

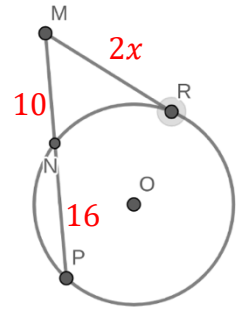
$$10x + 220 = 324$$

$$10x = 104$$

$$x = 10.4$$

$$(12 + (10.4)) + 10 = KP = 32.4$$

6. Circle O is shown with points N , R and P on the circle. Secant $\overline{MP}$ and tangent $\overline{MR}$ intersect at point M outside the circle. $PN = 16$, $MN = 10$ and $MR = 2x$. What is the value of x ? Round to the nearest thousandth.



$$10(26) = (2x)^2$$

$$4x^2 = 260$$

$$x^2 = 65$$

$$x \approx 8.062$$

7. Using the same image as question 6, Circle O is shown with points N , R and P on the circle. Secant $\overline{MP}$ and tangent $\overline{MR}$ intersect at point M outside the circle. $PN = x + 17$, $MN = 12$ and $MR = 24$. What is the value of x ?

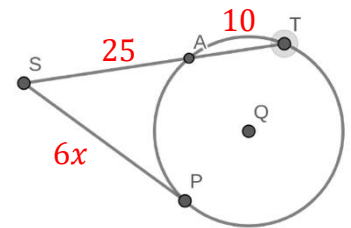
$$12(x + 17 + 12) = 24^2$$

$$12x + 348 = 576$$

$$12x = 228$$

$$x = 19$$

8. Circle Q is shown with points A , T and P on the circle. Secant $\overline{ST}$ and tangent $\overline{SP}$ intersect at point S outside the circle. $PS = 6x$, $AT = 10$ and $AS = 25$. What is the value of x ? Round to the nearest thousandth.



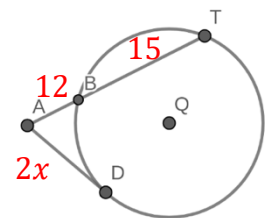
$$25(35) = (6x)^2$$

$$36x^2 = 875$$

$$x^2 \approx 24.306$$

$$x \approx 4.930$$

9. Daniella and Taylor are working on finding the length of $\overline{AD}$. Circle Q is shown with points B , T and D on the circle. Secant $\overline{AT}$ and tangent $\overline{AD}$ intersect at point A outside the circle. $BT = 15$, $AB = 12$ and $AD = 2x$. Their partial work solving for x is shown below. Whose work do you agree with? **Taylor's**



Daniella's work	Taylor's work
$(2x)^2 = 15(15 + 12)$	$(2x)^2 = 12(15 + 12)$
$4x^2 = 15(27)$	$4x^2 = 12(27)$
$4x^2 = 405$	$4x^2 = 324$
$x^2 = 101.25$	$x^2 = 81$

10. Using the image in Question 9, what is the length of $\overline{AD}$?

$$2(9) = AD = 18$$

$$x = \sqrt{81}$$

$$x = 9$$